10/14/24

<u>Example</u> Find LSS of

$$\underset{A}{\begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}} \underset{x}{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}} = \underset{b}{\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}}$$

Find the projection of b onto R(A)

Previous: Example $\quad b = s + \hat{s}, \quad s = \frac{1}{3}\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$

$$\hat{s} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{3}\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

$$= \dots$$

Now solve

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = s = \begin{bmatrix} 2/3 \\ 1/3 \\ 1/3 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 1 & 1 & 2/3 \\ 1 & 0 & 1/3 \\ 0 & 1 & 1/3 \end{array}\right] \xrightarrow{R_2 - R_1} \xrightarrow{R_3 - R_1} \dots$$

$$\hat{x} = \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 1/3 \end{bmatrix}$$ Least squares solution

<u>Note</u> What's the error in approximating b by $A\hat{x}$?

$$A\hat{x} = s \Rightarrow A\hat{x} = b - \hat{s} \Rightarrow \cancel{A\hat{x} = b} \; b - A\hat{x} = \hat{s}$$

Error: $\|\hat{s}\|_2$

Another approach follows

Preliminaries

Lemma 1 Let $A \in M_{n \times m}(\mathbb{R})$, $x \in \mathbb{R}^m$, $y \in \mathbb{R}^n$. Then

$$\langle Ax, y \rangle = \langle x, A^T y \rangle$$

Proof

$$\begin{aligned}
\langle Ax, y \rangle &= y^T A x \\
&= \cancel{(y^T A x)^T} \\
&= \cancel{x^T A^T y} \\
&= \cancel{(Ax)^T y} \\
&= \cancel{\langle y, A} \\
&= (A^T y)^T x \\
&= \langle x, A^T y \rangle
\end{aligned}$$

<u>Lemma 2</u> Let $A \in M_{n,m}(\mathbb{R})$. Then

$$\operatorname{rank}(A^T A) = \operatorname{rank}(A)$$

<u>Proof</u> By the "rank-nullity theorem" for every matrix X,

$$\underbrace{\operatorname{rank}(X)}_{\dim R(X)} + \dim \operatorname{Nul}(X) = \#\text{ of columns of } X$$

So to show this lemma, it is enough to show

$$A^T A x = 0 \iff A x = 0$$

(because: $N(A^T A) = N(A)$)

($\Leftarrow$) Let $Ax = 0$. Then $A^T A x = A^T 0 = 0$ ✓

($\Rightarrow$) Let $A^T A x = 0$. Then

$$\langle A^T A x, x \rangle = 0$$

$$\overset{\text{lemma}}{\Longrightarrow} \langle Ax, Ax \rangle = 0$$

$$\Longrightarrow \|Ax\|_2^2 = 0 \Longrightarrow Ax = 0 \quad \blacksquare$$

<u>Theorem</u> Let $A \in M_{n,m}(\mathbb{R})$, $b \in \mathbb{R}^n$.

Then $\exists x_0 \in \mathbb{R}^m$ such that

$$(A^T A) x_0 = A^T b.$$

Furthermore,

$$\| A x_0 - b \|_2 \leq \| A x - b \|_2 \quad \forall x \in \mathbb{R}^m,$$

that is, x_0 is the least-squares solution to $Ax = b$.

<u>Proof</u>

<u>Existence of x_0:</u>

Let $S := R(A)$. <u>Recall</u> that ~~the~~ an x_0 such that $Ax_0 \in S$ is the point closest to b (in $R(A)$) as long as

$$Ax_0 = \text{projection of } b \text{ onto } R(A)$$

So, to find x_0, write $b = s + \hat{s}$, $s \in S$, $\hat{s} \in S^\perp$

so

$$b = \underbrace{Ax_0}_{s} + \hat{s}$$

Thus $\hat{s} = b - Ax_0$

$\rightsquigarrow$

and so

$$\langle \underset{\text{arbitrary}}{x}, A^T A x_0 - A^T b \rangle = \langle x, A^T (A x_0 - b) \rangle$$

$$\overset{\text{lemma}}{=} \langle \underbrace{Ax}_{\in S}, \underbrace{A x_0 - b}_{\in S^\perp} \rangle$$

$$= 0$$

Because this holds for every $x \in \mathbb{R}^n$, we have that $A^T A x_0 - A^T b = 0$.

That is $A^T A x_0 = A^T b$ (existence)

<u>Also</u> x_0 must be the lss to $Ax = b$ because it minimizes $\|Ax - b\|_2$ ∎

PRACTICALLY we can find lss via previous theorem:

Solve for x_0 the equation

$$\boxed{A^T A x_0 = A^T b} \qquad x_0 = \text{lss. to } Ax = b$$

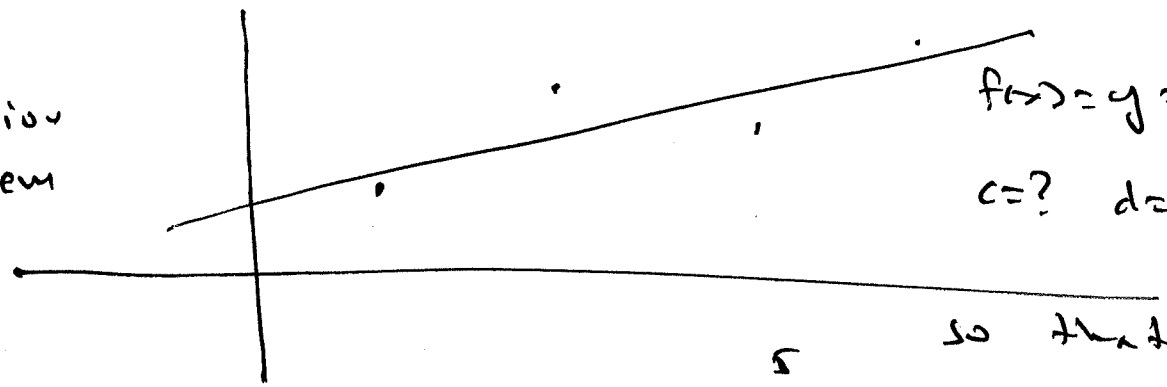
"normal equations"

<u>Note</u> Normal equations is the preferred method for finding lss because
- avoids orthonormal basis for $R(A)$
- $A^T A$ is pos. semidef.

Example

Find line of best-fit (minimize sum of squares of errors) that fits the points $(1,2)$, $(3,4)$, $(5,7)$, $(7,9)$, $(9,12)$

Linear Regression Problem



$f(x) = y = cx + d$

$c = ?$ $d = ?$

so that

$$\sum_{i=1}^{5} |f(x_i) - y_i|^2$$

is minimal

This means

$$Ax = \begin{bmatrix} 1 & 1 \\ 3 & 1 \\ 5 & 1 \\ 7 & 1 \\ 9 & 1 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 7 \\ 9 \\ 12 \end{bmatrix}$$

& we need the lsq of $Ax = b$

Normal equations

$$A^T A = \begin{bmatrix} 105 & 25 \\ 25 & 5 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 220 \\ 34 \end{bmatrix}$$

Solve

$$\begin{bmatrix} 105 & 25 \\ 25 & 5 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 220 \\ 34 \end{bmatrix}$$

$$\rightsquigarrow \ldots \quad \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 1.25 \\ .55 \end{bmatrix}$$

$$\boxed{y = 1.25x + .55}$$

10/16/24

Minimal Solutions

Task Given a consistent system

$$Ax = b,$$

find a solution s ($As = b$) such that

$$\|s\|_2 \leq \|u\|_2$$

for all u such that $Au = b$. We call this s a minimal solution to the system.

Theorem Let $A \in M_{n,m}(\mathbb{R})$, $b \in \mathbb{R}^n$ and suppose $Ax = b$ is consistent. Then,

(a) There exists a minimal solution s to this system, and $s \in R(A^T)$

(b) s is the only solution that lies in $R(A^T)$

i.e., $A(\underbrace{A^T u}_{\in R(A^T)}) = b \Rightarrow A^T u = s = \text{minimal}$

<u>Proof</u> Let $W := R(A^T)$

(a) Let $Ax = b$. Use ODT to write

$$x = s + \hat{s}, \quad s \in W, \quad \hat{s} \in W^\perp$$

Then

$$\begin{aligned}
\hat{s} \in W^\perp &\iff \hat{s} \perp A^T u, \ \forall u \in \mathbb{R}^n \\
&\iff \langle \hat{s}, A^T u \rangle = 0 \quad \text{—''—} \\
&\overset{\text{Lemma 1}}{\iff} \langle A\hat{s}, u \rangle = 0 \quad \text{—''—} \\
&\iff A\hat{s} = 0 \\
&\iff \hat{s} \in N(A)
\end{aligned}$$

We've thus proved the "Fundamental Theorem of Linear Algebra, part II":

$$(R(A^T))^\perp = N(A)$$

Thus, $\quad b = Ax = As + A\hat{s} = As$

That is $\underline{\underline{s}} = \text{proj}_W x$ is <u>a solution.</u>

If u is any solution, write

$$u = s_1 + \hat{s}_1, \quad s_1 \in W, \quad \hat{s}_1 \in W^\perp$$

Then

$$Au = b = As, \text{ i.e. } A(u-s) = 0$$

$$\text{i.e. } u - s \in N(A)$$

~~Notice $s_1 = s$ because~~

$$\cancel{s_1 + \hat{s}_1 - s \in N(A)}$$

So any two solution of $Ax = b$ differ by something in $N(A)$; that is,

$$u = s + v, \quad v \in N(A)$$

Hence

$$\|u\|_2 = \|s+v\|_2 \underset{\text{Pythag.}}{\overset{s \perp v}{=}} \|s\|_2 + \|v\|_2$$

$$\geq \|s\|_2$$

Thus, s is a <u>minimal</u> <u>solution</u>.

(b) Say $Aw = b$, $w \in W = R(A^T)$. Then

$$\begin{cases} A(w-s) = Aw - As = b - b = 0 \Rightarrow w - s \in N(A) \\ w - s \in W = R(A^T) \end{cases}$$

So $w - s \in W \cap W^\perp = \{0\}$

i.e. $w = s$. ∎

The above theorem suggests the following method for finding minimal solution:

<u>Step 1</u> Form & solve $AA^T u = b$ for u

<u>Step 2</u> Set $s := A^T u$. It is the minimal solution.

<u>Example</u> Find minimal solution of

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 2 & 0 & 2 \end{bmatrix}}_{A} x = \underbrace{\begin{bmatrix} -1 \\ 3 \end{bmatrix}}_{b}$$

It's consistent because $R(A) = \mathbb{R}^2$ but has infinite number of solutions

<u>Step 1</u> Form $AA^T = \dots = \begin{bmatrix} 3 & 4 \\ 4 & 8 \end{bmatrix}$

<u>Step 2</u> Solve. $\begin{bmatrix} 3 & 4 \\ 4 & 8 \end{bmatrix} \underbrace{\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}}_{u} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$

$\xrightarrow{1/3 R_1} \xrightarrow{R_2 - 4R_1} \dots \quad u_2 = \dfrac{26/9}{44/6}$ ✓

$u_1 = -\frac{1}{3} + \frac{4}{3} u_2$ ✓

$A^T u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ is guaranteed to be in $R(A^T)$

so $\boxed{s = A^T u}$ is the minimal solution

Remarks

1) Contrast

$A^T A \hat{x} = A^T b$ gives $\hat{x}$ = LSS of $Ax = b$

$AA^T u = b$ is used to find the minimal solution $s = A^T u$ of $Ax = b$

Of course, $A^T A \neq A A^T$ in general

2) $$R(A^T) = N(A)^\perp$$

Also true if you use $T, \perp$:

$$R(A)^\perp = N(A^T)$$

e.g.

$T, \perp$

$$R(A^T) = N(A)^\perp$$

10/18/24

To conclude: LSS / minimal solution

<u>Fact</u> One can combine least-squares & minimal solution ideas.

In our discussion of LSP, we had $\operatorname{rank} \underset{m\times n}{A} = m$, so $\operatorname{rank}(A^T A) = m$, i.e., there is a unique solution to LSP:

$$\hat{x} = (A^T A)^{-1} A^T b$$

But if we drop assumption on the rank of A, the normal equation will have several solutions. In that case, one can apply our method for finding the minimal solution of

$$\underbrace{(A^T A)}\, \hat{x} = A^T b$$

CHAPTER 4 : The Singular Value Decomposition (SVD)

Motivation

For symmetric $A \in M_n(\mathbb{R})$ we have "spectral theorem"

$$(*) \quad \boxed{A = \lambda_1 u_1 u_1^T + \lambda_2 u_2 u_2^T + \dots + \lambda_n u_n u_n^T}$$

where $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues of A (which are real) & u_j are corresponding eigenvectors : $Au_j = \lambda_j u_j$, $j = 1, 2, \dots, n$

(Recall: symmetric $\Rightarrow$ diagonalizable
$\Rightarrow$ n lin. indep. eigenvectors!)

Notice in $(*)$ A is written as a sum of rank-one matrices ($u_j u_j^T \in M_n(\mathbb{R})$, $\operatorname{rank} u_j u_j^T = 1$)

Natural question: Is there something like $(*)$ for arbitrary matrices?

Answer: YES

Theorem 4.1.1 (SVD Theorem)

Let $A \in M_{n,m}(\mathbb{R})$, $\operatorname{rank} A = r > 0$. Then

$$A = U \Sigma V^T$$

where $U \in M_n(\mathbb{R})$, $V = M_m(\mathbb{R})$ are orthogonal matrices, $\Sigma \in M_{n,m}(\mathbb{R})$ is "diagonal"

i.e.

$$\Sigma = \begin{bmatrix} \sigma_1 & & & & & \\ & \sigma_2 & & & 0 & \\ & & \ddots & & & \\ & & & \sigma_r & & \\ & 0 & & & 0 & \\ & & & & & 0 \\ & & & & & & \ddots \end{bmatrix}_{n \times m}$$

with

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$$

Proof & applications coming up ....

<u>Notes</u>

- U, V are <u>not</u> unique
- $\sigma_1, \sigma_2, \dots, \sigma_r$ are called the <u>singular values</u> of A
  They are the square roots of the nonzero eigenvalues of $A^T A$.
- Columns of U are know as the <u>left singular vectors</u> of A

- Columns of V are known as the <u>right singular vectors</u> of A

- Sometimes it will be convenient to introduce $\sigma_{r+1}, \sigma_{r+2}, \ldots \sigma_k$, $k = \min(m,n)$

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0 = \sigma_{r+1} = \cdots = \sigma_k$$

- By convention: 0 matrix has all singular values $= 0$

- Notice that we've generalizing Spectral Thm

$$\boxed{A = U\Sigma V^T = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \cdots + \sigma_r u_r v_r^T}$$

$$U = [u_1 | \cdots | u_m], \quad V = [v_1 | v_2 | \cdots v_n]$$

<u>Proof</u> (of SVD theorem)

Spectrum of $A^TA = \{\lambda_1, \lambda_2, \ldots, \lambda_m\}$

Since A^TA is positive semidefinite,

$$\lambda_j \geq 0 \quad , j = 1, 2, \ldots, m$$

Say $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_m \geq 0$

Since A^TA is symmetric (i.e., normal), it is orthogonally diagonalizable; that is $\exists\, v_1, v_2, \ldots, v_m$ mutually orthogonal eigenvectors of A^TA

$$(A^TA v_j = \lambda_j v_j).$$

<u>Claim</u> $\{Av_1, Av_2, \ldots, Av_m\}$ is also an orthogonal set, because

$$\begin{aligned}(Av_i)^T(Av_j) &= v_i^T A^T A v_j \\ &= v_i^T \lambda_j v_j \\ &= \lambda_j (v_i^T v_j) \\ &\overset{i \neq j}{=} \lambda_j (0) = 0.\end{aligned}$$

In fact, since $\operatorname{rank}(A) \overset{\text{lemma 2}}{=} \operatorname{rank}(A^T A) = r$,

we have that

$$\{ Av_1, Av_2, \ldots, Av_r \} \subseteq R(A)$$

must be an orthogonal basis for $R(A)$

(Prove $Av_j \neq 0$ for $j = 1, 2, \ldots, r$)

Let now $\{ u_1, u_2, \ldots, u_r \}$ comprise

$$u_j := \frac{Av_j}{\| Av_j \|_2}$$

So this set is an <u>orthonormal</u> basis for $R(A)$

Complete it to an orthonormal basis for $\mathbb{R}^n$

$$\{ u_1, u_2, \ldots, u_r, u_{r+1}, \ldots, u_n \}$$

and set $U = [u_1 | u_2 | \cdots | u_n]$: orthogonal.

TBC